Supplementary materials for "Coordinatewise Gaussianization: Theories and Applications"

November 3, 2021

S1 Additional Simulation Results

S1.1 A Visual Demonstration for Model D6

We visualize the simulation results for Model D6 in Section 5.1.3 to demonstrate the effect of the coordinatewise Gaussianization as the model deviates from normality. Note that, in Model D6, $(Y, \mathbf{X})$ is jointly normal when $\lambda = 1$, and we move away from normality as λ decreases. We plot the medians of $|\mathcal{M}|$ in Figure S1. We only plot the results for DCS and NS-DCS, as Wins-DCS is almost identical to NS-DCS. It can be seen that, when the normality assumption is severely violated at $\lambda = 0, 0.25$, NS-DCS outperforms DCS by a large margin. When $\lambda = 0.5, 0.75, 1$ and the model is close to normal, DCS is slightly better than NS-DCS.

S1.2 Comparison with More Features Screening Methods

For the models in Section 5.1.3, we also apply many popular feature screening methods for comparison. In particular, we include three robust screening methods as suggested by a referee: the sure independent ranking and screening (SIRS; Zhu et al. (2011)), the robust rank correlation screening (RRCS; Li et al. (2012)) and the quantile correlation screening (QCS; Ma & Zhang (2016)). These methods do not require moment conditions on the response and/or predictors to enjoy the SURE screening property. We further

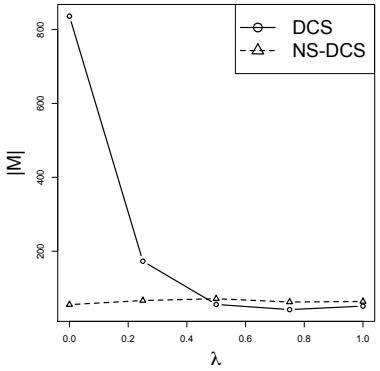


Figure S1: Medians of $|\mathcal{M}|$ for Model D6. We only plot the results for DCS and NS-DCS, as Wins-DCS is almost identical to NS-DCS.

consider the sure independence screening (SIS; Fan & Lv (2008)) and the nonparametric independence screening (NIS; Fan et al. (2011)) for a more comprehensive comparison. The results are summarized in Tables S1 and S2. Note that we omit Model D5, because none of these screening methods handle categorical responses.

Comparing Tables S1 and S2 with Table 2, we see that all the existing methods perform significantly worse than NS-DCS and Wins-DCS. This fact supports the application of Gaussianized distance correlation screening, even given the large number of existing screening methods.

S2 Proofs of Theorem 1 and related lemmas

Throughout the proofs, we make use of the following two functions

$$T_j^*(t) = T_j(t)\mathrm{I}(|T_j(t)| < \sqrt{2\log n}) + \mathrm{sign}(T_j(t)) \cdot \sqrt{2\log n} \cdot \mathrm{I}(|T_j(t)| \geq \sqrt{2\log n}) \quad (1)$$

$$F_j^*(t) = F_j(t)\mathrm{I}(|T_j(t)| < \sqrt{2\log n}) + \Phi\{\mathrm{sign}(T_j(t)) \cdot \sqrt{2\log n}\}\mathrm{I}(|T_j(t)| \geq \sqrt{2\log n}) \quad (2)$$

i.e., $T_j(t)$ Winsorized at $\pm\sqrt{2\log n}$ and $F_j(t)$ Winsorized at $\Phi(\pm\sqrt{2\log n})$. In addition, denote $\phi(z)$ as the probability density function for a standard normal random variable, and M, C as generic constants that can vary from place to place.

The following technical propositions and lemmas are important for our proofs. Propositions 1 & 2 are cited from the literature. We only present the proofs for Proposition 3 and Lemmas 12–4.

Table S1: The 5%, 25%, 50%, 75%, and 95% quantiles of the minimum model size $\mathcal{M}$ out of 200 replications for Models D1-D3, and the proportion of $\mathcal{P}_a$ with model size $d = 2[n/\log n]$. The numbers in the brackets are the true numbers of variables.

			$\mathcal{M}$					$\mathcal{P}_a$				$\mathcal{M}$					$\mathcal{P}_a$
Model		Method	5%	25%	50%	75%	95%		Model		Method	5%	25%	50%	75%	95%	
D1	(25)	SIS	4.0	4.0	5.0	7.0	26.2	0.98	D2		SIS	21.0	100.8	317.0	1182.0	3696.1	0.21
[4]		NIS	4.0	4.0	5.0	12.2	114.2	0.93	[2]		NIS	55.2	277.8	851.5	1788.2	3758.2	0.08
		SIRS	4.0	4.0	5.0	8.0	34.2	0.98			SIRS	2.0	2.0	2.0	9.0	209.7	0.88
		RRCS	4.0	4.0	4.0	6.0	27.4	0.99			RRCS	2.0	2.0	2.0	2.0	2.0	1.00
		QCS	4.0	4.0	5.0	7.0	28.0	0.98			QCS	2.0	2.0	2.0	2.0	44.2	0.97
	(26)	SIS	32.0	301.2	865.0	1456.8	1900.6	0.11	D3	Case 1:	SIS	287.5	652.0	1119.0	1560.8	1922.1	0.01
		NIS	4.0	9.0	39.0	130.5	547.6	0.63	[4]	$\epsilon \sim N(0,1)$	NIS	4.0	4.0	4.0	4.0	7.0	1.00
		SIRS	105.0	752.5	1279.5	1754.8	1958.2	0.04			SIRS	225.6	745.0	1205.5	1663.5	1925.4	0.02
		RRCS	136.2	663.2	1216.5	1705.0	1933.1	0.04			RRCS	255.5	706.2	1133.0	1536.2	1865.6	0.01
		QCS	99.8	642.5	1204.5	1659.8	1902.4	0.05			QCS	274.9	651.2	1188.5	1568.0	1874.2	0.01
	(27)	SIS	318.5	918.8	1377.5	1728.0	1922.5	0.00		Case 1:	SIS	398.6	1063.5	1472.0	1736.2	1963.4	0.01
		NIS	100.5	493.0	1130.5	1553.8	1857.8	0.04		$\epsilon \sim t(1)$	NIS	41.8	557.5	1182.5	1717.2	1936.0	0.07
		SIRS	202.5	747.0	1262.0	1661.8	1939.1	0.00			SIRS	273.6	718.5	1183.0	1569.5	1873.7	0.01
		RRCS	563.1	1107.5	1554.0	1811.8	1959.0	0.00			RRCS	306.9	677.2	1136.0	1599.5	1934.3	0.01
		QCS	508.0	1084.5	1494.0	1734.8	1966.0	0.00			QCS	260.4	743.0	1181.0	1593.2	1881.5	0.01
	(28)	SIS	136.6	473.0	1004.5	1601.0	1958.0	0.03		Case 2	SIS	42.6	253.2	469.5	721.8	944.2	0.14
		NIS	182.9	526.5	1013.5	1517.5	1851.3	0.01			NIS	11.0	152.2	514.0	856.0	989.0	0.22
		SIRS	62.1	257.8	654.5	1291.8	1818.5	0.07			SIRS	56.3	282.8	516.5	728.0	929.6	0.12
		RRCS	47.7	294.5	740.5	1325.8	1803.7	0.09			RRCS	39.9	247.2	477.0	704.0	914.1	0.14
		QCS	28.0	217.2	588.0	1191.5	1823.1	0.11			QCS	42.8	276.8	471.0	729.0	920.4	0.14

Table S2: The 5%, 25%, 50%, 75%, and 95% quantiles of the minimum model size $\mathcal{M}$ out of 200 replications for Model D4-D6, and the proportion of $\mathcal{P}_a$ with model size $d = 2[n/\log n]$. The numbers in the brackets are the true numbers of variables.

			$\mathcal{M}$					$\mathcal{P}_a$
Model		Method	5%	25%	50%	75%	95%	
D4		SIS	1071.0	1508.0	1718.0	1876.2	1972.2	0.00
[8]		NIS	1128.1	1509.5	1734.5	1869.2	1971.1	0.00
		SIRS	11.0	18.0	35.0	76.5	470.8	0.74
		RRCS	323.7	1033.0	1461.5	1797.5	1954.3	0.01
		QCS	454.0	1033.8	1493.0	1822.2	1965.3	0.01
D6	λ=0	SIS	230.8	623.8	928.5	1373.8	1808.4	0.00
[3]		SIRS	4.0	12.0	50.5	211.5	619.7	0.40
		RRCS	4.0	14.0	59.0	203.5	664.5	0.40
		QCS	3.0	9.0	33.0	127.5	551.0	0.50
	λ=0.25	SIS	34.0	159.0	391.0	738.5	1353.6	0.04
		NIS	83.8	326.8	629.5	973.5	1518.0	0.03
		SIRS	4.0	17.8	57.0	241.0	729.2	0.35
		RRCS	3.0	16.0	61.0	198.2	643.4	0.37
		QCS	3.0	10.8	42.0	139.2	611.3	0.44
	λ=0.5	SIS	4.0	17.8	68.5	208.5	849.4	0.32
		NIS	11.9	53.2	161.5	428.8	1185.3	0.19
		SIRS	5.0	13.8	67.0	184.0	556.5	0.38
		RRCS	4.0	16.5	70.0	190.2	563.1	0.39
		QCS	3.0	10.0	39.5	127.5	588.0	0.46
	λ=0.75	SIS	4.0	7.0	33.0	132.2	542.3	0.50
		NIS	3.0	12.0	52.5	166.8	750.1	0.43
		SIRS	4.0	15.0	60.5	249.0	615.4	0.36
		RRCS	3.0	14.5	53.5	230.0	606.7	0.39
		QCS	3.0	9.8	41.0	160.2	442.2	0.46
	λ=1	SIS	3.0	9.0	27.5	98.8	400.4	0.51
		NIS	3.0	9.8	36.0	122.5	534.4	0.49
		SIRS	4.0	15.8	56.0	207.5	509.2	0.40
		RRCS	3.0	16.0	68.0	201.5	488.0	0.38
		QCS	3.0	11.8	36.0	127.0	448.4	0.48

Proposition 1. *(McDiarmid 1989) Let $(X^1, \ldots, X^n)$ be independent random variables. Suppose a measurable function g satisfies $|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq c_i$, where $\mathbf{x}, \tilde{\mathbf{x}} \in \mathbb{R}^n$ only differ on the ith coordinate. Then for any $\epsilon > 0$, we have*

$$\Pr(|g(X^1, \ldots, X^n) - Eg(X^1, \ldots, X^n)| \geq \epsilon) \leq 2\exp(-2\epsilon^2 / \sum_i c_i^2)$$

Proposition 2. *(Abramovich et al. 2006, c.f Lemmas 12.1 & 12.3) For $z \geq 1$ and $0 < \eta \leq 0.01$, we have*

$$\phi(z)/(2z) \leq 1 - \Phi(z) \leq \phi(z)/z, \tag{3}$$

$$\Phi^{-1}(1 - \eta) \leq \sqrt{2\log \eta^{-1}}. \tag{4}$$

Proposition 3. *For any $t \in \mathbb{R}$, we have $|\widehat{T}_j^c(t)| \leq \sqrt{2\log(2(n+2))}$.*

Proof of Proposition 3. By definition,

$$|\widehat{T}_j^c(t)| \leq \max\{|\Phi^{-1}(\frac{n-c}{n-2c+1})|, |\Phi^{-1}(\frac{1-c}{n-2c+1})|)\} \leq \Phi^{-1}(1 - \frac{1}{2(n+2)}).$$

By (4), we have $\Phi^{-1}(1 - \frac{1}{2(n+2)}) \leq \sqrt{2\log(2(n+2))}$. □

Lemma 1. *For any positive integer n, we have $\frac{1}{n}\sum_{i=1}^{n-1}(\Phi^{-1})'(\frac{i}{n}) \leq 2\sqrt{2\log n} + 3\sqrt{2\pi}$.*

Proof of Lemma 12. For any x, we have $(\Phi^{-1})'(x) = \frac{1}{\phi(\Phi^{-1}(x))} = \sqrt{2\pi}\exp[\{\Phi^{-1}(x)\}^2/2]$. Therefore,

$$\frac{1}{n}\sum_{i=1}^{n-1}(\Phi^{-1})'(\frac{i}{n}) = \frac{1}{n}\sum_{i=1}^{n-1}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{i}{n})\}^2/2]$$

Because Φ^{-1} is symmetric around 0, we have

$$\begin{aligned}
&\frac{1}{n}\sum_{i=2}^{n-1}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{i}{n})\}^2/2] \leq \frac{2}{n}\sum_{i=\lceil n/2\rceil}^{n-1}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{i}{n})\}^2/2] + \frac{\sqrt{2\pi}}{n} \\
= &\frac{2}{n}\sum_{i=\lceil n/2\rceil}^{n-2}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{i}{n})\}^2/2] + \frac{2}{n}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{n-1}{n})\}^2/2] + \frac{\sqrt{2\pi}}{n} \\
\leq &\frac{2}{n}\sum_{i=\lceil n/2\rceil}^{n-2}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{i}{n})\}^2/2] + 2\sqrt{2\pi} + \frac{\sqrt{2\pi}}{n}
\end{aligned}$$

where the last inequality follows from Proposition 3. For the first term, note that $\exp(\Phi^{-1}(u)^2/2)$ is increasing in u for $u > 1/2$. It follows that

$$\begin{aligned}
&\frac{2}{n}\sum_{i=\lceil n/2\rceil}^{n-2}\sqrt{2\pi}\exp[\{\Phi^{-1}(\frac{i}{n})\}^2/2] \le 2\int_{1/2}^{\frac{n-1}{n}}\sqrt{2\pi}\exp[\{\Phi^{-1}(u)\}^2/2]du \\
= \;& 2\int_0^{\Phi^{-1}(\frac{n-1}{n})}\sqrt{2\pi}\exp(t^2/2)\phi(t)dt = 2\sqrt{2\pi}\Phi^{-1}(\frac{n-1}{n}) \le 4\sqrt{\pi\log n},
\end{aligned}$$

where we use (4). Therefore,

$$\frac{1}{n}\sum_{i=1}^{n-1}(\Phi^{-1})'(\frac{i}{n}) \le 2\sqrt{2\log n} + 2\sqrt{2\pi} + \frac{\sqrt{2\pi}}{n} \le 2\sqrt{2\log n} + 3\sqrt{2\pi}.$$

□

Lemma 2. *Suppose $Z, Z_1, \ldots, Z_n$ are i.i.d standard normal random variables.*

1. *(Bickel et al. 2008) There exists $\epsilon_0 > 0$ such that for any $0 < \epsilon < \epsilon_0$, we have*

$$\Pr(|\frac{1}{n}\sum_{i=1}^{n} Z_i^2 - 1| > \epsilon) \le C\exp(-Cn\epsilon^2). \tag{5}$$

2. *For any $\epsilon > 0$, we have*

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}|Z_i| - E|Z| \ge \epsilon) \le \exp(-\frac{n\epsilon^2}{2}). \tag{6}$$

   while, for $\epsilon \le \frac{4}{\sqrt{2\pi}}$, we have

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}|Z_i| - E|Z| < -\epsilon) \le \exp(-\frac{n\epsilon^2}{2}). \tag{7}$$

3. *For $n \ge 2$, we have*

$$\Pr(|Z| \ge \sqrt{2\log n}) \le \frac{n^{-1}}{\sqrt{\pi\log n}}. \tag{8}$$

4. *For $\epsilon \ge \frac{2n^{-1}}{\sqrt{\pi\log n}}$, we have*

$$\Pr\left\{\frac{1}{n}\sum_{i=1}^{n}1(|Z_i| \ge \sqrt{2\log n}) \ge \epsilon\right\} \le 2\exp(-n\epsilon^2/2) \tag{9}$$

Proof of Lemma 2. We first prove (6). Straightforward calculation shows that, for a standard normal random variable, we have

$$Ee^{t|Z|} = 2\Phi(t)\exp(\frac{t^2}{2})$$

When $t = 0$, we have $2\Phi(t) = \exp(tE|Z|)$. For $t > 0$, note that $E|Z| = \frac{2}{\sqrt{2\pi}}$ and hence

$$(2\Phi(t) - \exp(tE|Z|))' = 2\phi(t) - \frac{2}{\sqrt{2\pi}}\exp(\frac{2t}{\sqrt{2\pi}}) = \frac{2}{\sqrt{2\pi}}\{e^{-t^2/2} - \exp(\frac{2t}{\sqrt{2\pi}})\} < 0.$$

Therefore, $2\Phi(t) \leq \exp(tE|Z|)$ and it follows that

$$Ee^{t|Z|} \leq \exp(\frac{t^2}{2} + tE|Z|) \tag{10}$$

Hence, by Chernoff bound, we have (6).

For (7), again by straightforward calculation we have

$$Ee^{-t|Z|} = 2\Phi(-t)\exp(\frac{t^2}{2}).$$

For $t = 0$, we have $2\Phi(-t) = e^{-tE|Z|}$. For $t > 0$, we have

$$(2\Phi(-t) - e^{-tE|Z|})' = -2\phi(-t) + (E|Z|)e^{-tE|Z|} = \frac{2}{\sqrt{2\pi}}e^{-\frac{2t}{\sqrt{2\pi}}}(1 - e^{-\frac{t^2}{2}+\frac{2t}{\sqrt{2\pi}}}) \leq 0$$

when $0 < t < \frac{4}{\sqrt{2\pi}}$. Hence,

$$Ee^{-t|Z|} \leq \exp(\frac{t^2}{2} - tE|Z|)$$

for $0 < t < \frac{4}{\sqrt{2\pi}}$. Again by Chernoff bound, we have the desired conclusion.

Now we show (8). By (3), for $n > \sqrt{e}$ we have

$$\Pr(|Z| \geq \sqrt{2\log n}) \leq 2\frac{\phi(\sqrt{2\log n})}{\sqrt{2\log n}} = \frac{n^{-1}}{\sqrt{\pi\log n}}.$$

Finally, we show (9). By (8), for $\epsilon \geq \frac{2n^{-1}}{\sqrt{\pi\log n}}$, we have

$$\begin{aligned}
& \Pr\{\frac{1}{n}\sum_{i=1}^{n} 1(|Z_i| \geq \sqrt{2\log n}) \geq \epsilon\} \\
\leq\ & \Pr\{\frac{1}{n}\sum_{i=1}^{n} 1(|Z_i| \geq \sqrt{2\log n}) - E1(|Z| \geq \sqrt{2\log n}) \geq \epsilon/2\} \\
& + \Pr\{E1(|Z| \geq \sqrt{2\log n}) \geq \epsilon/2\} \\
=\ & \Pr\{\frac{1}{n}\sum_{i=1}^{n} 1(|Z_i| \geq \sqrt{2\log n}) - E1(|Z| \geq \sqrt{2\log n}) \geq \epsilon/2\} \\
\leq\ & 2\exp(-n\epsilon^2/2)
\end{aligned}$$

where the last inequality follows from Hoeffding's inequality. □

Lemma 3. *There exists a generic positive constant M such that*

$$E\{\sup_x |\widehat{F}_j(x) - F_j(x)|\} \leq M\sqrt{\frac{\log n}{n}} \tag{11}$$

$$E\{\sup_x |\widehat{F}_j(x) - F_j(x)| \mid X_j^i\} \leq M\sqrt{\frac{\log n}{n}} \tag{12}$$

Proof of Lemma 3. We first show (11). By the DKW inequality (Dvoretzky et al. 1956, Massart 1990) we have that, for any $\epsilon > 0$,

$$\begin{aligned} & E\sup_x |\widehat{F}_j(x) - F_j(x)| \\ \leq\ & \epsilon \Pr(\sup_x |\widehat{F}_j(x) - F_j(x)| \leq \epsilon) + 2\Pr(\sup_x |\widehat{F}_j(x) - F_j(x)| \geq \epsilon) \\ \leq\ & \epsilon + 4\exp(-2n\epsilon^2) \end{aligned}$$

Take $\epsilon = \sqrt{\frac{\log n}{n}}$ and we have

$$E\sup_x |\widehat{F}_j(x) - F_j(x)| \leq M\sqrt{\frac{\log n}{n}}.$$

Now we show (12). Define $\widehat{F}_j^{(-i)}(x)$ as the empirical CDF for X_j based on $(X_j^1, \ldots, X_j^{i-1}, X_j^{i+1}, \ldots, X_j^n)$. Then

$$\begin{aligned} & |\widehat{F}_j^{(-i)}(x) - \widehat{F}_j(x)| \\ =\ & |\frac{1}{n}\sum_{k=1}^n 1(X_j^k \leq x) - \frac{1}{n-1}\sum_{k\neq i} 1(X_j^k \leq x)| \\ \leq\ & |(\frac{1}{n} - \frac{1}{n-1})\sum_{k\neq i} 1(X_j^k \leq x)| + \frac{1}{n}1(X_j^i \leq x) \\ \leq\ & \frac{1}{n-1} + \frac{1}{n} = \frac{2}{n-1} \end{aligned} \tag{13}$$

Therefore, by (13) and (11), we have

$$\begin{aligned} & E\{\sup_x |\widehat{F}_j(x) - F_j(x)| \mid X_j^i\} \\ \leq\ & E\{\sup_x |\widehat{F}_j(x) - \widehat{F}^{(-i)}(x)| \mid X_j^i\} + E\{\sup_x |\widehat{F}^{(-i)}(x) - F_j(x)| \mid X_j^i\} \\ \leq\ & \frac{2}{n-1} + E\{\sup_x |\widehat{F}_j^{(-i)}(x) - F_j(x)|\} \\ \leq\ & \frac{2}{n-1} + M\sqrt{\frac{\log(n-1)}{n-1}} \leq M\sqrt{\frac{\log n}{n}}. \end{aligned}$$

□

Lemma 4. *For F_j^* defined in* (2)*, we have* $E\left\{\frac{1}{n}\sum_{i=1}^n(\Phi^{-1})'(F_j^*(X_j^i))\right\} \le 2\sqrt{2\log n}+1$.

Proof of Lemma 4. Note that

$$\begin{aligned}
& E(\Phi^{-1})'(F_j^*(X_j^i)) \\
= & \int_{-\sqrt{2\log n}}^{\sqrt{2\log n}} \sqrt{2\pi}\exp(z^2/2)\cdot\phi(z)dz + 2\sqrt{2\pi}\exp(2\log n/2)\int_{\sqrt{2\log n}}^{\infty}\phi(z)dz \\
= & 2\sqrt{2\log n} + 2\sqrt{2\pi}n(1-\Phi(\sqrt{2\log n})) \\
\le & 2\sqrt{2\log n} + 2\sqrt{2\pi}n\frac{1}{2\sqrt{\pi\log n}}\exp(-\log n) \\
= & 2\sqrt{2\log n}+1,
\end{aligned}$$

and we have the conclusion. □

To show Theorem 1, we note that

$$\frac{1}{n}\sum_{i=1}^n|\widehat{T}_j(X_j^i)-T_j(X_j^i)| \le \frac{1}{n}\sum_{i=1}^n|\widehat{T}_j(X_j^i)-T_j^*(X_j^i)| + \frac{1}{n}\sum_{i=1}^n|T_j^*(X_j^i)-T_j(X_j^i)|$$

We have the following results concerning these two terms.

Lemma 5. *For* $\frac{8n^{-1}}{\sqrt{\pi\log n}} < \epsilon < 1$*, we have*

$$\Pr(\frac{1}{n}\sum_{i=1}^n|T_j^*(X_j^i)-T_j(X_j^i)|\ge\epsilon) \le C\exp(-\frac{Cn\epsilon^2}{2}).$$

Lemma 6. *Set* $g(X_j^1,\ldots,X_j^n)=\frac{1}{n}\sum_{i=1}^n|\widehat{T}_j(X_j^i)-T_j^*(X_j^i)|$*. Then for any* $\epsilon>0$*, we have*

$$\Pr(|g(X_j^1,\ldots,X_j^n)-Eg(X_j^1,\ldots,X_j^n)|\ge\epsilon) \le C\exp(-C\frac{n\epsilon^2}{\log n}).$$

Lemma 7. *Set* $g(X_j^1,\ldots,X_j^n)=\frac{1}{n}\sum_{i=1}^n|\widehat{T}_j(X_j^i)-T_j^*(X_j^i)|$*. We have* $|Eg(X_j^1,\ldots,X_j^n)| \le C\frac{\log n}{\sqrt{n}}$.

Lemma 8. *For* $\epsilon > C\frac{\log n}{\sqrt{n}}$*, we have*

$$\Pr(\frac{1}{n}\sum_{i=1}^n|\widehat{T}_j(X_j^i)-T_j^*(X_j^i)|\ge\epsilon) \le C\exp(-C\frac{n\epsilon^2}{\log n}).$$

Proof of Lemma 19. By the definition of T_j^*, we have

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^n\sum_{i=1}^n |T_j^*(X_j^i) - T_j(X_j^i)| = \frac{1}{n}\sum_{i=1}^n |T_j(X_j^i) 1\{|T_j(X_j^i)| \geq \sqrt{2\log n}\} \\
\leq\ & \sqrt{\frac{1}{n}\sum_{i=1}^n \{T_j(X_j^i)\}^2} \cdot \sqrt{\frac{1}{n}\sum_{i=1}^n 1\{|T_j(X_j^i)| \geq \sqrt{2\log n}\}}
\end{aligned}$$

It follows that

$$\begin{aligned}
&\Pr(\frac{1}{n}\sum_{i=1}^n\sum_{i=1}^n |T_j^*(X_j^i) - T_j(X_j^i)| \geq \epsilon) \\
\leq\ & \Pr(\sqrt{\frac{1}{n}\sum_{i=1}^n \{T_j(X_j^i)\}^2} \cdot \sqrt{\frac{1}{n}\sum_{i=1}^n 1\{|T_j(X_j^i)| \geq \sqrt{2\log n}\}} \geq \epsilon) \\
\leq\ & \Pr(\sqrt{\frac{1}{n}\sum_{i=1}^n \{T_j(X_j^i)\}^2} > 2) + \Pr(\sqrt{\frac{1}{n}\sum_{i=1}^n \{T_j(X_j^i)\}^2} \leq 2, \frac{1}{n}\sum_{i=1}^n 1\{|T_j(X_j^i)| \geq \sqrt{2\log n}\} \geq \frac{\epsilon}{2}) \\
=\ & \Pr(\frac{1}{n}\sum_{i=1}^n \{T_j(X_j^i)\}^2 - 1 > 3) + \Pr(\frac{1}{n}\sum_{i=1}^n 1\{|T_j(X_j^i)| \geq \sqrt{2\log n}\} \geq \frac{\epsilon^2}{4}) \\
\leq\ & \Pr(\frac{1}{n}\sum_{i=1}^n \{T_j(X_j^i)\}^2 - E\{T_j(X_j^i)\}^2 > \epsilon) + \Pr(\frac{1}{n}\sum_{i=1}^n 1\{|T_j(X_j^i)| \geq \sqrt{2\log n}\} \geq \frac{\epsilon}{4}) \\
\leq\ & C\exp(-Cn\epsilon^2),
\end{aligned}$$

where the last inequality follows from (5) & (9). □

Proof of Lemma 14. We prove our conclusion by first showing that, for any $\mathbf{x} = (x_1, x_2, \ldots, x_n), \tilde{\mathbf{x}} = (\tilde{x}_1, x_2, \ldots, x_n)$, we have $|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \leq \frac{C\sqrt{\log n}}{n}$, where $\mathbf{x}$ and $\tilde{\mathbf{x}}$ only differ on ith coordinate.

Without loss of generality, assume that $\mathbf{x}$ and $\tilde{\mathbf{x}}$ only differ on the first coordinate. Let $\tilde{F}_j(x) = \frac{1}{n}\sum_{i=2}^n 1(x_i \leq x) + \frac{1}{n}1(\tilde{x}_1 \leq x)$ and $\tilde{T}_j(x) = \Phi^{-1}\left(\frac{n\tilde{F}_j(x)}{n+1}\right)$. Then

$$\begin{aligned}
|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| &= |\frac{1}{n}\sum_{i=1}^n |\widehat{T}_j(x_i) - T_j^*(x_i)| - \frac{1}{n}\sum_{i=2}^n |\tilde{T}_j(x_i) - T_j^*(x_i)| - \frac{1}{n}|\tilde{T}_j(\tilde{x}_1) - T_j^*(\tilde{x}_j^1)|| \\
&\leq \frac{1}{n}\sum_{i=2}^n |\widehat{T}_j(x_i) - \tilde{T}_j(x_i)| + \frac{1}{n}|\widehat{T}_j(x_1) - \tilde{T}_j(\tilde{x}_1)| + \frac{1}{n}|T_j^*(\tilde{x}_1) - T_j^*(x_1)|
\end{aligned}$$

By Proposition 3, the last two terms are no greater than $\frac{2\sqrt{2\log 2(n+2)}}{n}$. For the first term, set R_j^i to be the rank of x_i in $\mathbf{x}$, $\tilde{R}_j^1$ to be the rank of x_1 in $\tilde{\mathbf{x}}$, and $\tilde{R}_j^i$ to be the rank of x_i in $\tilde{\mathbf{x}}$ for $j > 1$. Without loss of

generality, assume that $1 \le R_j^1 = l < \tilde{R}_j^1 = m \le n$. Then $\tilde{R}_j^i \le R_j^i \le \tilde{R}_j^i + 1$ for $j > 1$. Consequently,

$$\begin{aligned} & \frac{1}{n}\sum_{i=2}^{n} |\widehat{T}(x_j^i) - \tilde{T}(x_j^i)| = \frac{1}{n}\sum_{i=l}^{\min\{m,n-1\}} |\Phi^{-1}(\frac{i+1}{n+1}) - \Phi^{-1}(\frac{i}{n+1})| \\ \le & \frac{1}{n}\sum_{i=1}^{n-1} |\Phi^{-1}(\frac{i+1}{n+1}) - \Phi^{-1}(\frac{i}{n+1})| = \frac{1}{n}\sum_{i=1}^{n-1} (\Phi^{-1})'(\xi_i)(\frac{i+1}{n+1} - \frac{i}{n+1}), \end{aligned}$$

where $\xi_i \in [\frac{i}{n+1}, \frac{i+1}{n+1}]$. Therefore, $\sum_{i=1}^{n-1}(\Phi^{-1})'(\xi_i)(\frac{i+1}{n+1} - \frac{i}{n+1}) = \frac{1}{n+1}\sum_{i=1}^{n-1}(\Phi^{-1})'(\xi_i)$.

Note that $(\Phi^{-1})'(\xi_i) = \sqrt{2\pi}\exp[\{\Phi^{-1}(\xi_i)\}^2/2]$ is an increasing function of $|\Phi^{-1}(\xi_i)|$. Because $\xi_i \in [\frac{i}{n+1}, \frac{i+1}{n+1}]$, we must have that $\Phi^{-1}(\frac{i}{n+1}) \le \Phi^{-1}(\xi_i) \le \Phi^{-1}(\frac{i+1}{n+1})$, which implies that $|\Phi^{-1}(\xi_i)| \le \max\{|\Phi^{-1}(\frac{i}{n+1})|, |\Phi^{-1}(\frac{i+1}{n+1})|\}$. Consequently, $(\Phi^{-1})'(\xi_i) \le (\Phi^{-1})'(\frac{i}{n+1}) + (\Phi^{-1})'(\frac{i+1}{n+1})$, and

$$\begin{aligned} & \sum_{i=1}^{n-1}(\Phi^{-1})'(\xi_i) \le \frac{1}{n}\sum_{i=1}^{n-1}(\Phi^{-1})'(\frac{i+1}{n+1}) + \frac{1}{n+1}\sum_{i=1}^{n-1}(\Phi^{-1})'(\frac{i}{n+1}) \\ \le & 2(\frac{1}{n+1}\sum_{i=1}^{n}(\Phi^{-1})'(\frac{i}{n+1})) \le 2(2\sqrt{2\log(n+1)} + 3\sqrt{2\pi}) + \frac{2\sqrt{2\log(2(n+2))}}{n}, \end{aligned}$$

where the last inequality follows from Lemma 12. Consequently,

$$|g(\mathbf{x}) - g(\tilde{\mathbf{x}})| \le \frac{2(2\sqrt{2\log(n+1)} + 3\sqrt{2\pi})}{n} + \frac{2\sqrt{2\log\{2(n+2)\}}}{n^2} + \frac{2\sqrt{2\log\{2(n+2)\}}}{n}.$$

By Proposition 1, we have the desired conclusion. □

Proof of Lemma 7. Note that

$$\begin{aligned} & g(X_j^1, \ldots, X_j^n) = \frac{1}{n}\sum_{i=1}^{n} |\Phi^{-1}\left(\frac{n\widehat{F}_j(X_j^i)}{n+1}\right) - \Phi^{-1} \circ F_j^*(X_j^i)| \\ = & \frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i)|\frac{n\widehat{F}_j(X_j^i)}{n+1} - F_j^*(X_j^i)| \le \frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i)\sup_x|\frac{n\widehat{F}_j(x)}{n+1} - F_j^*(x)| \\ \le & \frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i)(\sup_x|\frac{n\widehat{F}_j(x)}{n+1} - \widehat{F}_j(x)| + \sup_x|\widehat{F}_j(x) - F_j(x)| + \sup_x|F_j(x) - F_j^*(x)|) \\ \le & \frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i)(\frac{1}{n+1} + \sup_x|\widehat{F}_j(x) - F_j(x)| + \sup_x|F_j(x) - F_j^*(x)|) \end{aligned}$$

where ξ_i is between $\frac{n\widehat{F}_j(X_j^i)}{n+1}$ and $F_j^*(X_j^i)$. By Proposition 2, $\sup_x|F_j(x) - F_j^*(x)| \le 1 - \Phi(\sqrt{2\log n}) \le \frac{n^{-1}}{2\sqrt{\pi\log n}}$. On the other hand,

$$\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i) \le \frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\frac{i}{n+1}) + \frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(F_j^*(X_j^i))$$

Therefore, by Lemmas 12 & 4, we have

$$\begin{aligned} E\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i) &\leq E\{\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\frac{i}{n+1})+\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(F_j^*(X_j^i))\} \\ &\leq 2\sqrt{2\log(n+1)}+3\sqrt{2\pi}+2\sqrt{2\log n}+1 \end{aligned}$$

It follows that $g(X_j^1,\ldots,X_j^n)\leq \dfrac{C\sqrt{\log n}}{n}+E\{\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\xi_i)\sup_x|\widehat{F}_j(x)-F_j(x)|\}$. Now

$$\begin{aligned} & E\frac{1}{n}\sum_{R_j^i\neq n}(\Phi^{-1})'(\xi_i)\sup_x|\widehat{F}_j(x)-F_j(x)| \\ \leq\ & E\left[\{\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(\frac{i}{n+1})+\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(F_j^*(X_j^i))\}\sup_x|\widehat{F}_j(x)-F_j(x)|\right] \end{aligned}$$

Combining Lemma 12 and (11), we have

$$E\{\frac{1}{n}\sum_{i=1}^{n-1}(\Phi^{-1})'(\frac{i}{n})\sup_x|\widehat{F}_j(x)-F_j(x)|\}\leq(C\sqrt{\log n})\cdot E\sup_x|\widehat{F}_j(x)-F_j(x)|\leq C\frac{\log n}{\sqrt{n}}.$$

On the other hand, combine (12) and Lemma 4 and we have

$$\begin{aligned} & E\{\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(F_j^*(X_j^i))\sup_x|\widehat{F}_j(x)-F_j(x)|\} \\ =\ & E\left[\{\frac{1}{n}\sum_{i=1}^{n}E\{(\Phi^{-1})'(F_j^*(X_j^i))\sup_x|\widehat{F}_j(x)-F_j(x)|\mid X_j^i\}\right] \\ =\ & E\left[\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(F_j^*(X_j^i))E\{\sup_x|\widehat{F}_j(x)-F_j(x)|\mid X_j^i\}\right] \\ \leq\ & C\frac{\sqrt{\log n}}{\sqrt{n}}E(\Phi^{-1})'(F_j^*(X_j^i))\leq C\frac{\log n}{\sqrt{n}}. \end{aligned}$$

Therefore,

$$E\{\frac{1}{n}\sum_{i=1}^{n}(\Phi^{-1})'(F_j^*(X_j^i))\sup_x|\widehat{F}_j(x)-F_j(x)|\}\leq C\frac{\log n}{\sqrt{n}},\quad |Eg(X_j^1,\ldots,X_j^n)|\leq C\frac{\log n}{\sqrt{n}}.$$

The conclusion follows. □

Proof of Lemma 13. Lemma 13 is a direct consequence of the Lemmas 14 & 7. □

With these results, we can easily show Theorems 1.

Proof of Theorem 1. Lemmas 19 & 13 imply Equation (5). By the union bound argument we have (6).

□

S3 Proofs for the Gaussian copula Model

Lemma 9. *For any* $\epsilon > M\frac{n^{-1/2}}{\sqrt[4]{\pi \log n}}$, *we have*

$$\Pr(|\frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)T_k^*(X_k^i) - \sigma_{jk}| \geq \epsilon) \leq C\exp(-C\frac{n\epsilon^2}{\log^2 n})$$

Proof of Lemma 9. First, note that

$$\begin{aligned}
& |E(T_j^*(X_j^i)T_k^*(X_k^i)) - \sigma_{jk}| = |E\{(T_j^*(X_j^i)T_k^*(X_k^i)) - (T_j(X_j^i)T_k(X_k^i))\}| \\
= \; & |E\{T_j^*(X_j^i)T_k^*(X_k^i) - T_j(X_j^i)T_k(X_k^i)\}1(\cup_{l=j,k}\{|T_l(X_l^i)| \geq \sqrt{2\log n}\}) \\
\leq \; & \sqrt{E\{T_j^*(X_j^i)T_k^*(X_k^i) - T_j(X_j^i)T_k(X_k^i)\}^2} \cdot \sqrt{\Pr(\cup_{l=j,k}\{|T_l(X_l^i)| \geq \sqrt{2\log n}\})} \\
\leq \; & \sqrt{E\{T_j^2(X_j^i)T_k^2(X_k^i)\}\{\Pr(|T_j(X_j^i)| \geq \sqrt{2\log n}) + \Pr(|T_k(X_k^i)| \geq \sqrt{2\log n})\}} \\
\leq \; & \sqrt[4]{E\{T_j^4(X_j^i)\}E\{T_k^4(X_k^i)\}} \cdot M\frac{n^{-1/2}}{\sqrt[4]{\pi \log n}} \qquad (14) \\
\leq \; & M\frac{n^{-1/2}}{\sqrt[4]{\pi \log n}},
\end{aligned}$$

where we use (8) to obtain (14). On the other hand, note that $|T_j^*(X_j^i)T_k^*(X_k^i)| \leq 4\log n$. By Hoeffding's inequality, we have

$$\Pr(|\frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)T_k^*(X_k^i) - E\{T_j^*(X_j^i)T_k^*(X_k^i)\}| \geq \epsilon/2) \leq C\exp(-C\frac{n\epsilon^2}{\log^2 n})$$

By noting that

$$\begin{aligned}
& \Pr(|\frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)T_k^*(X_k^i) - \sigma_{jk}| \geq \epsilon) \\
\leq \; & \Pr(|\frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)T_k^*(X_k^i) - E\{T_j^*(X_j^i)T_k^*(X_k^i)\}| \geq \epsilon/2) \\
& + \Pr(|E(T_j^*(X_j^i)T_k^*(X_k^i)) - \sigma_{jk}| \geq \epsilon/2)
\end{aligned}$$

we have the desired conclusion. □

Proof of Theorem 3. Note that

$$\begin{aligned}
& |\widehat{\sigma}_{jk} - \sigma_{jk}| \\
\leq \; & |\frac{1}{n}\sum_{i=1}^{n} \widehat{T}_j(X_j^i)\widehat{T}_k(X_k^i) - \frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)T_k^*(X_k^i)| + |\frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)T_k^*(X_k^i) - \sigma_{jk}| \\
\equiv \; & L_1 + L_2
\end{aligned}$$

Now

$$\begin{aligned} L_1 &\leq |\frac{1}{n}\sum_{i=1}^{n}(\widehat{T}_j(X_j^i) - T_j^*(X_j^i))\widehat{T}_k(X_k^i)| + |\frac{1}{n}\sum_{i=1}^{n} T_j^*(X_j^i)(\widehat{T}_k(X_k^i) - T_k^*(X_k^i))| \\ &\leq C\sqrt{\log n}(\frac{1}{n}\sum_{i=1}^{n}|\widehat{T}_j(X_j^i) - T_j^*(X_j^i)|) + C\sqrt{\log n}(\frac{1}{n}\sum_{i=1}^{n}|\widehat{T}_k(X_k^i) - T_k^*(X_k^i)|) \end{aligned}$$

By Lemmas 13, we have that $\Pr(L_1 \geq \epsilon/2) \leq C\exp(-\frac{Cn\epsilon^2}{\log^2 n})$. By Lemma 9, we have that $\Pr(L_2 \geq \epsilon/2) \leq C\exp(-C\frac{n\epsilon^2}{\log^2 n})$. Hence, we have the desired conclusion.

□

Proof of Theorem 4. Combine Theorem 3 with the proofs in Xue & Zou (2012) and we have the desired conclusions. □

S4 Proofs for nearest shrunken centroids classifier

Proof of Lemma 2. Note that

$$\begin{aligned} & \Pr(\widehat{\mathcal{D}} = \mathcal{D}) \leq \Pr(\min_{j\in\mathcal{D}}|\tilde{\mu}_{yj}| > \max_{j\notin\mathcal{D}}|\tilde{\mu}_{yj}|) && (15) \\ = \; & \Pr(\min_{j\in\mathcal{D}}|\tilde{\mu}_{yj}| > \max_{j\notin\mathcal{D}}|\tilde{\mu}_{yj}|, \min_{j\in\mathcal{D}}|\tilde{\mu}_{yj}| > 2\delta) \\ & + \Pr(\min_{j\in\mathcal{D}}|\tilde{\mu}_{j}| > \max_{j\notin\mathcal{D}}|\tilde{\mu}_{yj}|, \min_{j\in\mathcal{D}}|\tilde{\mu}_{yj}| \leq 2\delta) && (16) \\ \leq \; & \Pr(\min_{j\in\mathcal{D}}|\tilde{\mu}_{yj}| > 2\delta) + \Pr(\max_{j\notin\mathcal{D}}|\tilde{\mu}_{yj}| \leq 2\delta) && (17) \\ \leq \; & \Pr(\min_{j\in\mathcal{D}}|\bar{\epsilon}_{yj}| > \delta) + \Pr(\max_{j\notin\mathcal{D}}|\bar{\epsilon}_{yj}| \leq 2\delta) && (18) \\ & && (19) \end{aligned}$$

where $\bar{\epsilon}_{yj}$ is the sample mean of ϵ_j within Class y. Because ϵ_j follows the Cauchy distribution, so does $\bar{\epsilon}_{yj}$. Denote $G_w = \Pr(W \geq w)$ for W following the Cauchy distribution. Then we have that

$$\Pr(\min_{j\in\mathcal{D}}|\bar{\epsilon}_{yj}| > \delta) + \Pr(\max_{j\notin\mathcal{D}}|\bar{\epsilon}_{yj}| \leq 2\delta) = G_\delta^d + (1 - 2G_{2\delta})^{p-d} \to 0 \quad (20)$$

as long as $d \to \infty$ and $p - d \to \infty$. □

Proof of Proposition 1. Without loss of generality, assume that, for any $j \in \mathcal{D}$, $\mu_{+j} > \mu_{-j}$. We have

$$T_j(\mu_{+j} + \epsilon_j) - T_j(\mu_{-j} + \epsilon_j) > 0 \quad (21)$$

everywhere, because T_j is strictly monotone. It follows that

$$E(T_j(X_j) \mid Y = +1) \neq E(T_j(X_j) \mid Y = -1). \tag{22}$$

□

To prove Theorems 5 & 6, we need the following Lemmas 10 & 11.

Lemma 10. *For n iid random variables $W^i \in \mathbb{R}$ with mean zero, if there exists $M > 0$ and a positive integer k such that $E(W^i)^l \leq M^l$ for all $l \leq 2k$, we must have that*

$$E\left[\left(\sum_{i=1}^{n} W^i\right)^{2k}\right] \leq n^k k^{2k} M^{2k}, \tag{23}$$

$$\Pr(|\frac{1}{n}\sum_{i=1}^{n} W^i| \geq \epsilon) \leq \frac{k^{2k}M^{2k}}{(\sqrt{n}\epsilon)^{2k}} \text{ for any } \epsilon > 0. \tag{24}$$

Lemma 11. *For random variables $Z^i, U^i, i = 1, \ldots, n$, where $Z^i \sim N(0,1)$ and $|U^i| \leq 1$, define $W^i = Z^i U^i$. We have that, for any $0 < \epsilon < \frac{1}{\sqrt{2}}$,*

$$\Pr(|\frac{1}{n}\sum_{i=1}^{n} W^i - EW^i| \geq \epsilon) \leq 2\exp(-\frac{n\epsilon^2}{4}). \tag{25}$$

The proofs of Lemmas 10 & 11 involve two technical propositions.

Proposition 4. *[Birgé et al. (1998, c.f. Lemma 8)] Let $X_1, \ldots, X_n$ be independent real valued random variables. Assume that there exists some positive numbers v, c such that*

$$\frac{1}{n}\sum_{i=1}^{n} E[|X_i|^m] \leq \frac{m!}{2} v^2 c^{m-2} \text{ for all } m \geq 2. \tag{26}$$

Then for any positive $\epsilon > 0$ and $S_n = \sum_{i=1}^{n} X_i$,

$$\Pr\{S_n - ES_n \geq n\epsilon\} \leq \exp\left(\frac{-n\epsilon^2/2}{v^2 + c\epsilon}\right) \tag{27}$$

Proposition 5. *For a standard normal random variable Z, we have $E|Z|^m \leq \frac{m!}{2}(\sqrt{2})^{m-2}$.*

Proof of Proposition 5. Straightforward calculation shows that

$$E|Z|^m = \begin{cases} 2^{(m-1)/2}(\frac{m-1}{2})! \cdot \dfrac{2}{\sqrt{2\pi}} & \text{for } m \text{ odd} \\ \dfrac{m!}{2^{m/2}(\frac{m}{2})!} & \text{for } m \text{ even} \end{cases} \tag{28}$$

It is easy to verify that, for odd m, $E|Z|^m \leq \frac{m!}{2} \cdot \sqrt{\frac{2}{\pi}}(\sqrt{2})^{m-2}$; for even m, $E|Z|^m \leq \frac{m!}{2}(\frac{1}{\sqrt{2}})^{m-2}$, and the conclusion follows. □

With Propositions 4 & (5), we present the proofs of Lemmas 10 & 11.

Proof of Lemma 10. We first show (23). Because W^i are independent and have zero means,

$$E\left[\left(\sum_{i=1}^{n} W^i\right)^{2k}\right] = E\left[\sum_{k_1+\cdots+k_n=2k} \binom{2k}{k_1,\cdots,k_n} \prod_{i=1}^{n} (W^i)^{k_i}\right] \tag{29}$$

$$= \sum_{k_1+\cdots+k_n=2k} \binom{2k}{k_1,\cdots,k_n} \prod_{i=1}^{n} E\left[(W^i)^{k_i}\right] \tag{30}$$

$$\leq \left\{\sum_{(k_1,\cdots,k_n)\in\mathcal{K}} \binom{2k}{k_1,\cdots,k_n}\right\} \cdot \max_{k\in\mathcal{K}} \prod_{i=1}^{n} E\left[(W^i)^{k_i}\right], \tag{31}$$

where $\mathcal{K} = \{(k_1,\ldots,k_n) : k_1+\cdots+k_n = 2k, \text{ and } k_1,\cdots,k_n \text{ are all even}\}$. By the proof of Lemma 2 in Ravikumar et al. (2011),

$$\sum_{(k_1,\cdots,k_n)\in\mathcal{K}} \binom{2k}{k_1,\cdots,k_n} \leq n^k k^{2k} \tag{32}$$

By the assumption that $E(W^i)^l \leq M^l$ for all $l \leq 2k$, we have that $\max_{k\in\mathcal{K}} \prod_{i=1}^{n} E\left[(W^i)^{k_i}\right] < M^{2k}$. And (23) follows.

For (24), by Markov's inequality,

$$\Pr(|\frac{1}{n}\sum_{i=1}^{n} W^i| \geq \epsilon) = \Pr((\sum_{i=1}^{n} W^i)^{2k} \geq (n\epsilon)^{2k}) \leq \frac{E(\sum_{i=1}^{n} W^i)^{2k}}{(n\epsilon)^{2k}} \leq \frac{n^k k^{2k} M^{2k}}{(n\epsilon)^{2k}} \tag{33}$$

where the last inequality follows from (23). And the conclusion follows. □

Proof of Lemma 11. Because $Z_i U_i \leq |Z_i|$, we have $E|Z_i U_i|^m \leq E|Z_i|^m$. Combine Proposition 4 with Lemma 5 and we have

$$\Pr(\frac{1}{n}\sum_{i=1}^{n} W^i - EW^i > \epsilon) \leq \exp(-\frac{n\epsilon^2/2}{1+\sqrt{2}\epsilon}) \leq \exp(-\frac{n\epsilon^2}{4}) \tag{34}$$

for $0 < \epsilon < \frac{1}{\sqrt{2}}$. Repeat the above proof for $-W^i$ and we have the desired conclusion. □

Now we are ready to prove Theorems 5 & 6.

Proof of Theorem 5. We first show the first conclusion. Note that $\widehat{\mathcal{D}} = \mathcal{D}$ is implied by $\sqrt{n}|\tilde{\mu}_{yj} - \mu_{yj}| < \Delta$ for all y, j. By Lemma 10, we have that

$$\Pr(\sqrt{n}|\tilde{\mu}_{yj} - \mu_{yj}| < \Delta \text{ for all } y, j) = \Pr(|\frac{1}{n_y}\sum_{Y^i=y}\epsilon_j^i| < \Delta/\sqrt{n} \text{ for all } y, j) \geq 1 - C\exp(-Cn\delta^2) \tag{35}$$

and the first conclusion follows. The second conclusion can be shown similarly by noting that ϵ_j is sub-Gaussian (Buldygin & Kozachenko 2000). □

Proof of Theorem 6. Define $\widehat{\eta}_{yj}^{(\mathrm{o})} = \frac{1}{n_y}\sum_{Y^i=y} T_j(X_j^i)$. Note that $\widehat{\eta}_{yj}^{(\mathrm{o})} = \frac{1}{n}\sum_{i=1}^n T_j(X_j^i)1(Y^i = y)$. By Lemma 11, we have that

$$\Pr(\sqrt{n}|\widehat{\eta}_{yj}^{(\mathrm{o})} - \eta_{yj}| \geq \Delta) \leq C\exp(-Cn\epsilon^2).$$

The first conclusion follows. The second conclusion can be shown in the same way by applying Theorem 1.

The first conclusion can be shown in the same way as Theorem 5 by applying Lemma 11. The third conclusion can be shown in the same way as Theorem 5 by apply Lemma 11 and Theorem 1. □

S5 Proofs for Distance Correlation Screening

Proof of Lemma 3. Because ω_k is the distance correlation between $T_{X_k}(X_k)$ and $T_Y(Y)$, it has to be between 0 and 1. We turn to the proof of the second conclusion. Under the bivariate normal model, where the dependence between X_k and Y can be fully characterized by their Pearson correlation $\rho_k = \mathrm{Cor}(X_k, Y)$, it is known that DC is a strictly monotone function of $|\rho_k|$ (Székely et al. 2007). We further notice that, under the bivariate normality, we have $T_k(X_k) = \dfrac{X_k - E(X_k)}{\sqrt{\mathrm{Var}(X_k)}}$ and $T_Y(Y) = \dfrac{Y - E(Y)}{\sqrt{\mathrm{Var}(Y)}}$. Hence, we must have that GDC equals DC, and GDC continues to be a strictly monotone function of $|\rho_k|$. □

The following two technical lemmas are useful for our proofs.

Lemma 12. *For independent standard normal random variables $Z^i, \ldots, Z^n$, and $\epsilon > \dfrac{2n^{-1}}{\sqrt{\pi \log n}}$, we have*

$$\Pr(\frac{1}{n}\sum_{l=1}^n (Z^i)^2 1(|Z^i| \geq \sqrt{2\log n}) \geq \epsilon) \leq C\exp(-Cn\epsilon^2)$$

Proof of Lemma 12.

$$\begin{aligned}
& \frac{1}{n}\sum_{l=1}^{n}(Z^i)^2 1(|Z^i| \geq \sqrt{2\log n}) \\
\leq \quad & \frac{1}{n}\sum_{l=1}^{n}\{(Z^i)^2 - E((Z^i)^2)\} \cdot 1(|Z^i| \geq \sqrt{2\log n}) \\
+ \quad & E(Z^i)^2 \cdot \frac{1}{n}\sum_{i=1}^{n} 1(|Z^i| \geq \sqrt{2\log n}) \\
\leq \quad & \frac{1}{n}\sum_{l=1}^{n}\{(Z^i)^2 - E((Z^i)^2)\} + \frac{1}{n}\sum_{i=1}^{n} 1(|Z^i| \geq \sqrt{2\log n})
\end{aligned}$$

By the Chernoff bound, we have

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}\{(Z^i)^2 - E((Z^i)^2)\} \geq \frac{\epsilon}{2}) \leq \exp(-Cn\epsilon^2)$$

On the other hand, a straightforward calculation shows that

$$\Pr(|Z^i| \geq \sqrt{2\log n}) \leq \frac{n^{-1}}{\sqrt{\pi \log n}}$$

Combine this fact with the Hoeffding's inequality and we have that, for $\epsilon > \dfrac{2n^{-1}}{\sqrt{\pi \log n}}$,

$$\Pr(\frac{1}{n}\sum_{i=1}^{n} 1(|Z^i| \geq \sqrt{2\log n}) \geq \frac{\epsilon}{2}) \leq \exp(-Cn\epsilon^2)$$

And the conclusion follows. □

Lemma 13. *For independent pairs of random variables $(Z_1^i, Z_2^i)_{i=1}^n$, where all the pairs are independent of each other and $Z_k^i \sim N(0,1)$ for $k = 1, 2$, define $\tilde{Z}_k^i$ as Z_k^i Winsorized at $\pm\sqrt{2\log n}$. For $\epsilon > \dfrac{2n^{-1}}{\sqrt{\pi \log n}}$, we have*

$$\Pr(\frac{1}{n}\sum_{i=1}^{n} |\tilde{Z}_1^i - Z_1^i||\tilde{Z}_2^i - Z_2^i| \geq \epsilon) \leq C\exp(-Cn\epsilon^2) \tag{36}$$

$$\Pr(\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} |\tilde{Z}_1^i - Z_1^i||\tilde{Z}_2^j - Z_2^j| \geq \epsilon) \leq C\exp(-Cn\epsilon^2) \tag{37}$$

Proof of Lemma 13. We only show Equation (37). Equation (36) can be proved in the same way. For

(37), we have

$$\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}|\tilde{Z}_1^i - Z_1^i||\tilde{Z}_2^j - Z_2^j| \le \frac{1}{2n}\sum_{i=1}^{n}(\tilde{Z}_1^i - Z_1^i)^2 + \frac{1}{2n}\sum_{i=1}^{n}(\tilde{Z}_2^i - Z_2^i)^2$$
$$\le \sum_{k=1,2}\left\{\frac{1}{n}\sum_{i=1}^{n}(Z_k^i)^2 1(|Z_k^i| > \sqrt{2\log n})\right\}$$

And the conclusion follows from Lemma 12. □

Lemma 14. *There exists a constant $M > 0$ such that, for $\epsilon > \frac{M\log n}{\sqrt{n}}$, we have*

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}\|\hat{T}_{X_k}(X_k^i) - T_X(X_k^i)\|^2 \ge \epsilon) \le C\exp(-C\frac{n\epsilon^2}{\log^2 n}) \quad (38)$$

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}\|\hat{T}_{Y}(Y^i) - T_Y(Y^i)\|^2 \ge \epsilon) \le C\exp(-C\frac{n\epsilon^2}{\log^2 n}) \quad (39)$$

Proof of Lemma 14. We only show Equation (38). Equation (39) can be shown in the same way. For (38), note that

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n}(\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i))^2 \\
\le\ & \frac{1}{n}\sum_{i=1}^{n}|\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)|\cdot(|\hat{T}_{X_{k,j}}(X_{k,j}^i) - T^*_{X_{k,j}}(X_{k,j}^i)| + |T^*_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)|) \\
\le\ & C\frac{\sqrt{\log n}}{n}\sum_{i=1}^{n}|\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)| \\
&+\frac{1}{n}\sum_{i=1}^{n}|\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)|\cdot|T^*_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)| \\
\le\ & C\cdot\frac{\sqrt{\log n}}{n}\sum_{i=1}^{n}|\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)| \\
&+\frac{1}{2}\cdot\frac{1}{n}\sum_{i=1}^{n}(\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i))^2 + \frac{1}{2}\cdot\frac{1}{n}\sum_{i=1}^{n}(T^*_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i))^2
\end{aligned}$$

The second inequality follows from Proposition 3. Combine the last inequality with Lemma 1 and 12 and we have

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}(\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i))^2 \ge \epsilon) \le C\exp(-C\frac{n\epsilon^2}{\log^2 n})$$

The desired conclusion follows from the fact that $\frac{1}{n}\sum_{i=1}^{n}\|\hat{T}_{X_k}(X_k^i) - T_{X_k}(X_k^i)\|^2 = \sum_{j=1}^{p_k}\{\frac{1}{n}\sum_{i=1}^{n}(\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i))^2\}$. □

Now we turn to the proof of Theorem 7. Let $\mathbf{T} = (T_{X_k}, T_Y)$. Define $T^*_{X_{k,j}}$ as $T_{X_{k,j}}$ Winsorized at $\pm\sqrt{2\log n}$. Similarly, we have T^*_Y. Recall the definitions of $S_j(X,Y), j = 1,2,3$ in (16)–(18). We introduce the shorthand notation

$$\begin{aligned} S_j(\mathbf{T}) &= S_j(T_{X_k}(X_k), T_Y(Y)) \\ S_j(\hat{\mathbf{T}}) &= S_j(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) \end{aligned}$$

for $j = 1,2,3$.

We first derive some basic results for $\hat{T}_{X_k}$ and $\hat{T}_Y$.

Lemma 15. *1. For any $t \in \mathbb{R}^{p_k}, s \in \mathbb{R}^q$, we have*

$$\begin{aligned} \|\hat{T}_X(t)\| &\le \sqrt{2p\log n}, \\ \|\hat{T}_Y(s)\| &\le \sqrt{2q\log n}. \end{aligned}$$

2. There exists $M > 0$, such that, for $\epsilon \ge \dfrac{M\sqrt{\log n}}{\sqrt{n}}$, we have

$$\begin{aligned} \Pr(\frac{1}{n}\sum_{i=1}^n \|\hat{T}_{X_k}(X_k^i) - T_{X_k}(X_k^i)\| \ge \epsilon) &\le Cp_k \exp(-C\frac{n\epsilon^2}{p_k^3 \log n}) \\ \Pr(\frac{1}{n}\sum_{i=1}^n \|\hat{T}_Y(Y^i) - T_y(Y^i)\| \ge \epsilon) &\le Cq \exp(-C\frac{n\epsilon^2}{q^3 \log n}) \end{aligned}$$

Proof of Lemma 15. For a vector $U \in R^d$, note that

$$\|U\| \le \sqrt{d} \max_{j=1,\cdots,d} |U_j|. \tag{40}$$

The first conclusion immediately follows from the original propositions. For the second conclusion, note that

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n \|\hat{T}_{X_k}(X_k^i) - T_{X_k}(X_k^i)\| &\le \frac{1}{n}\sum_{i=1}^n \sum_{i=1}^n \sqrt{p_k} \max_j |\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)| && (41) \\ &\le \frac{1}{n}\sum_{i=1}^n \sum_{i=1}^n \sqrt{p_k} \sum_j |\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)| && (42) \\ &= \sqrt{p_k}\sum_{j=1}^{p_k} \{\frac{1}{n}\sum_{i=1}^n |\hat{T}_{X_{k,j}}(X_{k,j}^i) - T_{X_{k,j}}(X_{k,j}^i)|\} && (43) \end{aligned}$$

It follows that

$$\Pr(\frac{1}{n}\sum_{i=1}^{n}\|\hat{T}_{X_k}(X_k^i)-T_{X_k}(X_k^i)\|\geq\epsilon) \tag{44}$$

$$\leq \Pr(\sqrt{p_k}\sum_{j=1}^{p_k}\{\frac{1}{n}\sum_{i=1}^{n}|\hat{T}_{X_{k,j}}(X_{k,j}^i)-T_{X_{k,j}}(X_{k,j}^i)|\}\geq\epsilon/\sqrt{p_k}) \tag{45}$$

$$\leq \Pr(\max_j\frac{1}{n}\sum_{i=1}^{n}|\hat{T}_{X_{k,j}}(X_{k,j}^i)-T_{X_{k,j}}(X_{k,j}^i)|\}\geq\epsilon/{p_k}^{3/2}) \tag{46}$$

$$\leq Cp_k\exp(-\frac{n\epsilon^2}{p_k^3\log n}). \tag{47}$$

□

Recall that we assume that q is fixed and p_k are uniformly bounded above. Under this assumption, they do not affect the convergence rate. Therefore, in the rest of the proof we no longer keep track of them.

Lemma 16. *There exists a constant $M>0$ such that, for $\epsilon>\frac{M\log n}{\sqrt{n}}$, we have*

$$\Pr(|S_1(\hat{\mathbf{T}})-S_1(\mathbf{T})|\geq\epsilon)\leq C\exp(-C\frac{n\epsilon^2}{\log^2 n})$$

Proof of Lemma 16. First, note that

$$\begin{aligned}
&S_{1j}(\hat{\mathbf{T}})-S_{1j}(\mathbf{T})\\
=\;&\frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}\Big\{\|\hat{T}_{X_k}(X_k^l)-\hat{T}_{X_k}(X_k^m)\|\cdot\|\hat{T}_Y(Y^l)-\hat{T}_Y(Y^m)\|\\
&-\|T_{X_k}(X_k^l)-T_{X_k}(X_k^m)\|\cdot\|T_Y(Y^l)-T_Y(Y^m)\|\Big\}\\
=\;&\frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}\Big\{\big(\|\hat{T}_{X_k}(X_k^l)-\hat{T}_{X_k}(X_k^m)\|-\|T_{X_k}(X_k^l)-T_{X_k}(X_k^m)\|\big)\cdot\|\hat{T}_Y(Y^l)-\hat{T}_Y(Y^m)\|\\
&+\|T_{X_k}(X_k^l)-T_{X_k}(X_k^m)\|\cdot\big(\|\hat{T}_Y(Y^l)-\hat{T}_Y(Y^m)\|-\|T_Y(Y^l)-T_Y(Y^m)\|\big)\Big\}\\
\leq\;&\frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}\Big\{\big(\|\hat{T}_{X_k}(X_k^l)-T_{X_k}(X_k^l)\|+\|\hat{T}_{X_k}(X_k^m)-T_{X_k}(X_k^m)\|\big)\cdot\|\hat{T}_Y(Y^l)-\hat{T}_Y(Y^m)\|\Big\}\\
&+\frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}\Big\{\big(\|\hat{T}_Y(Y^l)-T_Y(Y^l)\|+\|\hat{T}_Y(Y^m)-T_Y(Y^m)\|\big)\cdot\|T_{X_k}(X_k^l)-T_{X_k}(X_k^m)\|\Big\}
\end{aligned}$$

The above inequality follows from the Minkowski inequality. Since

$$\|\hat{T}_Y(Y^l) - \hat{T}_Y(Y^m)\| \leq \|\hat{T}_Y(Y^l)\| + \|\hat{T}_Y(Y^m)\|$$

and

$$\begin{aligned}\|T_{X_k}(X_k^l) - T_{X_k}(X_k^m)\| \leq & \|T_{X_k}(X_k^l) - \hat{T}_{X_k}(X_k^l)\| + \|\hat{T}_{X_k}(X_k^l)\| \\ & + \|T_{X_k}(X_k^m) - \hat{T}_{X_k}(X_k^m)\| + \|\hat{T}_{X_k}(X_k^m)\|,\end{aligned}$$

by the first part of Lemma 15, we have

$$\begin{aligned}& S_{1j}(\hat{\mathbf{T}}) - S_{1j}(\mathbf{T}) \\ \leq\ & C\frac{\sqrt{q\log n}}{n}\sum_{l=1}^{n}\|\hat{T}_{X_k}(X_k^l) - T_{X_k}(X_k^l)\| + C\frac{\sqrt{p\log n}}{n}\sum_{l=1}^{n}\|\hat{T}_Y(Y^l) - T_Y(Y^l)\| \\ & + \frac{C}{n}\sum_{l=1}^{n}\|\hat{T}_{X_k}(X_k^l) - T_{X_k}(X_k^l)\|^2 + \frac{C}{n}\sum_{l=1}^{n}\|\hat{T}_Y(Y^l) - T_Y(Y^l)\|^2\end{aligned}$$

By the second part of Lemmas 15 and Lemma 14 we have the desired conclusion.

□

Lemma 17. *There exists a constant $M > 0$ such that, for $\epsilon > \dfrac{M\log n}{\sqrt{n}}$, we have*

$$\Pr(|S_2(\hat{\mathbf{T}}) - S_2(\mathbf{T})| \geq \epsilon) \leq C\exp(-C\frac{n\epsilon^2}{\log^2 n})$$

Proof of Lemma 17. Using the same technique as that in the proof of Lemma 16, we have

$$\begin{aligned}& S_2(\hat{\mathbf{T}}) - S_2(\mathbf{T}) \\ =\ & \frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}(\|\hat{T}_{X_k}(X_k^l) - \hat{T}_{X_k}(X_k^m)\| - \|T_{X_k}(X_k^l) - T_{X_k}(X_k^m)\|) \\ & \cdot\frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}\|\hat{T}_Y(Y^l) - \hat{T}_Y(Y^m)\| \\ & + \frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}(\|T_{X_k}(X_k^l) - T_{X_k}(X_k^m)\|) \\ & \cdot\frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n}(\|\hat{T}_Y(Y^l) - \hat{T}_Y(Y^m)\| - \|T_Y(Y^l) - T_Y(Y^m)\|) \\ \leq\ & C\cdot\frac{1}{n}\sum_{l=1}^{n}\|\hat{T}_{X_k}(X_k^l) - T_{X_k}(X_k^l)\|\cdot\frac{1}{n}\sum_{l=1}^{n}\|\hat{T}_Y(Y^l) - T_Y(Y^l)\| \\ & + C\cdot\frac{\sqrt{\log n}}{n}\sum_{l=1}^{n}\|\hat{T}_{X_k}(X_k^l) - T_{X_k}(X_k^l)\| + C\cdot\frac{\sqrt{\log n}}{n}\sum_{l=1}^{n}\|\hat{T}_Y(Y^l) - T_Y(Y^l)\|\end{aligned}$$

By the second part of Lemma 15 we have the desired conclusion. □

Lemma 18. *There exists a constant $M > 0$ such that, for $\epsilon \geq \frac{M \log n}{\sqrt{n}}$, we have*

$$\Pr(|S_3(\hat{\mathbf{T}}) - S_3(\mathbf{T})| \geq \epsilon) \leq C \exp(-C \frac{n\epsilon^2}{\log^2 n})$$

Lemma 18 can be proven in a similar way as Lemma 16 and we omit the proof here.

Lemma 19. *For $b > 0$, $|a| \leq b$ and $0 < \epsilon \leq \frac{3b^2}{4}$, we have*

1. $$\frac{a+\epsilon}{\sqrt{b^2-\epsilon}} - \frac{a}{b} \leq (1 + \frac{1}{b})\epsilon$$

2. $$\frac{a}{b} - \frac{a-\epsilon}{\sqrt{b^2+\epsilon}} \leq (\frac{1}{2} + \frac{1}{b})\epsilon$$

Proof. For the first conclusion, note that

$$\begin{aligned} \frac{a+\epsilon}{\sqrt{b^2-\epsilon}} - \frac{a}{b} &= \frac{(a+\epsilon)b - a\sqrt{b^2-\epsilon}}{b\sqrt{b^2-\epsilon}} \leq \frac{2\{(a+\epsilon)b - a\sqrt{b^2-\epsilon}\}}{b^2} \\ &\leq \frac{2|a|b(1 - \sqrt{1 - \frac{\epsilon}{b^2}}) + b\epsilon}{b^2} \leq \frac{2\epsilon}{1 + \sqrt{1 - \frac{\epsilon}{b^2}}} + \frac{\epsilon}{b} \\ &\leq (1 + \frac{1}{b})\epsilon \end{aligned}$$

For the second conclusion, note that

$$\frac{a}{b} - \frac{a-\epsilon}{\sqrt{b^2+\epsilon}} = \frac{a\sqrt{b^2+\epsilon} - (a-\epsilon)b}{\sqrt{b^2+\epsilon}b} \leq \frac{b\epsilon + |a|(\sqrt{b^2+\epsilon} - b)}{b^2} \leq (\frac{1}{2} + \frac{1}{b})\epsilon$$

□

Lemma 20. 1. *For $\epsilon \geq \frac{M \log n}{\sqrt{n}}$, we have*

$$\Pr(|\widehat{\mathrm{dcov}}(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) - \widehat{\mathrm{dcov}}(T_{X_k}(X_k), T_Y(Y))| \geq \epsilon) \leq C \exp(-C \frac{n\epsilon^2}{\log^2 n})$$

2. *For $\epsilon \geq \frac{M \log n}{\sqrt{n}}$, we have*

$$\Pr(|\widehat{\mathrm{dcor}}(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) - \widehat{\mathrm{dcor}}(T_{X_k}(X_k), T_Y(Y))| \geq \epsilon) \leq C \exp(-C \frac{n\epsilon^2}{\log^2 n})$$

Proof of Lemma 20. The first conclusion is a direct consequence of Lemmas 16, 17 and 18. Combine the first conclusion with Lemma 19 and we have the desired conclusion. □

Proof of Theorem 7. Note that

$$\begin{aligned}
&\Pr(|\widehat{\mathrm{dcor}}(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) - \mathrm{dcor}(T_{X_k}(X_k), T_Y(Y))| \geq \epsilon) \\
\leq &\Pr(|\widehat{\mathrm{dcor}}(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) - \widehat{\mathrm{dcor}}(T_{X_k}(X_k), T_Y(Y))| \geq \frac{\epsilon}{2}) \\
&+ \Pr(|\widehat{\mathrm{dcor}}(T_{X_k}(X_k), T_Y(Y)) - \mathrm{dcor}(T_{X_k}(X_k), T_Y(Y))| \geq \frac{\epsilon}{2})
\end{aligned}$$

Let $\epsilon = Mn^{-\kappa}$, apply Lemma 20 and Proposition 1 and we have that

$$\begin{aligned}
&\Pr(|\widehat{\mathrm{dcor}}(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) - \mathrm{dcor}(T_{X_k}(X_k), T_Y(Y))| \geq Mn^{-\kappa}) \\
\leq &\Pr(|\widehat{\mathrm{dcor}}(\hat{T}_{X_k}(X_k), \hat{T}_Y(Y)) - \widehat{\mathrm{dcor}}(T_{X_k}(X_k), T_Y(Y))| \geq \frac{M \log n}{2n^{\kappa+\gamma}}) \\
&+ \Pr(|\widehat{\mathrm{dcor}}(T_{X_k}(X_k), T_Y(Y)) - \mathrm{dcor}(T_{X_k}(X_k), T_Y(Y))| \geq \frac{Mn^{-\kappa}}{2}) \\
\leq &C[\exp(-Cn^{1-2(\kappa+\gamma)}) + n\exp(-Cn^{\gamma})]
\end{aligned}$$

□

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